

Fig. 2: NEC transmission code (Image courtesy: www.vishay.com)

SOME IR RECEIVER MODULES FOR DIFFERENT CARRIER FREQUENCIES

Module	Frequency	Module	Frequency
TSOP1730	30kHz	TSOP1733	33kHz
TSOP1736	36kHz	TSOP1737	36.7kHz
TSOP1738	38kHz	TSOP1740	40kHz
TSOP1756	56kHz	—	—

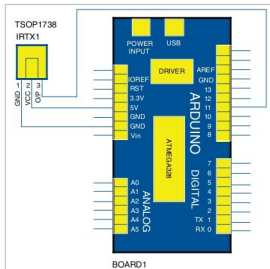


Fig. 3: Circuit diagram of Arduino based Windows PC volume remote control

transmission of address 00110111 and command data 00011010 is performed by sending bits as:

"00110111" "11001000" "00011010" "11100101"

In a special version of NEC code, the pre-burst, including all address and data bits, is repeated in each

Circuit diagram of Arduino PC volume control is shown in Fig. 3. Hardware components required are Arduino Uno, TSOP1738 IR receiver, breadboard and some jumper wires.

Working of the project is fairly simple. Arduino Uno decodes the

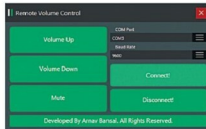


Fig. 4: Screenshot of Remote Volume Control software

108ms time slot for as long as the key is pressed.

IR receiver modules.

The most suitable IR receivers for receiving NEC code are those with AGC4 setting as these have the best noise suppression while still supporting this data format. Some examples are TSOP4438, TSOP58438 and TSOP75438. Receiver modules for different carrier frequencies are given in the Table above.

Circuit and working

key of the remote and then compares it with predefined variables to figure out which key is pressed. If it finds a match, it sends the signal to the PC, which is interpreted by Remote Volume Control software (designed by author), which further performs the required task. Screenshot of Remote

Volume Control software is shown in Fig. 4. The most important step is to configure the desired remote control device with Arduino Uno.

Software

The source codes (IR_Key_Test.ino and IR_Final.ino) are written in Arduino programming language. The software used to program Arduino Uno is Arduino IDE, which can be downloaded free of cost from www.arduino.cc/en/Main/Software

Arduino, which is an open source software, makes it easy to write and upload the code to the board. It runs on Windows, Mac OS X and Linux. The environment is written in Java and based on Processing and other open source software.

The library used in this project is IRremote library. It allows Arduino Uno to decode IR signals coming from the remote. To install the library, copy IRremote folder and paste it in the